程序控制

Process Control

Chapter 1: Introduction
Chapter 2: Modeling Tools for Process Dynamics

中英文講稿 Bilingual Lecture Notes

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Unit 1: 為什麼需要程序控制？

Unit 1: Why Process Control?

在化工製程中，程序控制扮演著不可或缺的角色。以下是五大核心原因：

In chemical processes, process control plays an indispensable role. Here are the five key reasons:

1. 提升製程安全性 (Enhanced Process Safety)

即時監測溫度、壓力、流量等參數，防止失控反應、爆炸或有毒物質洩漏。

Real-time monitoring of temperature, pressure, flow rate, etc. to prevent runaway reactions, explosions, or toxic leaks.

2. 滿足環保法規 (Satisfying Environmental Constraints)

精確控制排放物，確保製程符合日益嚴格的環保法規，避免罰款與停工。

Precisely control emissions to ensure compliance with increasingly strict environmental regulations, avoiding fines and shutdowns.

3. 嚴格產品品質規格 (Meeting Product Quality Specs)

精密控制反應條件，確保產品純度、組成與物性符合品質要求。

Precisely control reaction conditions to ensure product purity, composition, and properties meet quality requirements.

4. 更高效利用原料與能源 (Efficient Use of Resources)

最佳化操作條件，減少浪費與能耗，降低單位生產成本。

Optimize operating conditions, reduce waste and energy consumption, lower unit production costs.

5. 提高獲利能力 (Increased Profitability)

減少廢品、提高效率、降低能耗 → 整體提升經濟效益。

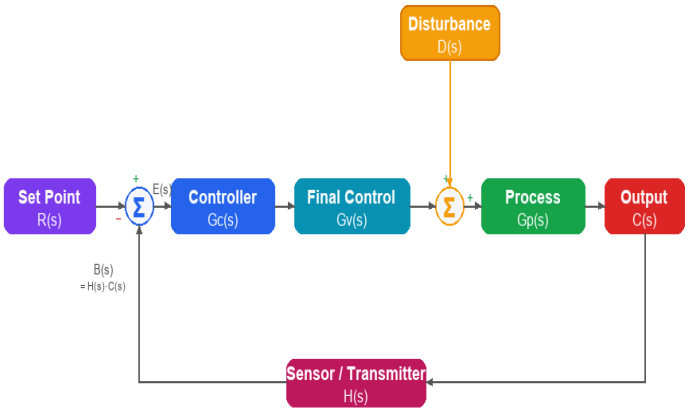
Reduce waste, improve efficiency, lower energy use → overall economic benefit improvement.

Unit 2: 控制系統方塊圖

Unit 2: Control System Block Diagram

任何控制系統都可以用方塊圖 (Block Diagram) 表示。閉迴路 (Closed-loop) 控制系統包含以下元件：

Any control system can be represented by a Block Diagram. A closed-loop control system contains the following components:



閉迴路控制系統方塊圖 / Closed-Loop Control System Block Diagram

Set Point R(s) — 設定值

操作人員設定的目標值（如反應器 80°C）。控制系統目標：使輸出趨向設定值。

The desired target value set by the operator (e.g., reactor at 80°C). Control objective: drive output toward set point.

Comparator (Σ) — 比較器（+ − 節點）

計算誤差 E(s) = R(s) − B(s)。一加一減，正偏差 = 輸出低於設定值。

Computes error E(s) = R(s) − B(s). One +, one −. Positive error = output below set point.

Controller Gc(s) — 控制器

根據誤差 E(s) 計算控制動作。類型：P、PI、PID。決定校正力道。

Computes control action based on error E(s). Types: P, PI, PID. Determines corrective effort.

Final Control Element Gv(s) — 最終控制元件

執行控制命令的裝置，如控制閥調節流量、變頻器調節幫浦轉速。

Device that executes control commands, e.g., control valve adjusting flow, VFD adjusting pump speed.

Additive Summing Junction (Σ+) — 加法節點（+ + 節點）

此節點將控制動作 Gv(s) 的輸出與干擾 D(s) 相加（皆為 +），進入製程。注意與比較器不同：比較器是 R(s)−B(s)（一加一減），此處是純加法。

This junction adds the output of Gv(s) and the disturbance D(s) together (both +) before entering the process. Unlike the comparator which computes R(s)−B(s), this is pure addition.

Disturbance D(s) — 干擾

不受控的外部因素：進料溫度波動、環境變化、原料濃度變動等。

Uncontrolled external factors: feed temperature fluctuations, ambient changes, raw material concentration variations.

Process Gp(s) — 製程

被控制的物理/化學程序（反應器、蒸餾塔、熱交換器等）。

The physical/chemical process being controlled (reactor, distillation column, heat exchanger, etc.).

Output C(s) — 受控變數

要控制的變數：溫度、濃度、液位等。目標：C(s) → R(s)。

The variable to be controlled: temperature, concentration, level, etc. Goal: C(s) → R(s).

Sensor / Transmitter H(s) — 感測器

量測輸出變數（熱電偶、流量計等）並回傳 B(s) = H(s)·C(s)。

Measures the output variable (thermocouple, flowmeter, etc.) and feeds back B(s) = H(s)·C(s).

⚠ 為什麼 B(s) ≠ C(s)？為什麼需要 H(s)？
C(s) 是實際的物理量（如溫度 80°C），但控制器讀不懂溫度。感測器 H(s) 負責兩件事：
1. 信號轉換：把物理量轉成控制器能讀的電子信號（如 4–20 mA、0–5 V）。H(s) 包含這個轉換比例（增益）。
2. 感測器動態：感測器有反應時間（如熱電偶有熱慣性），量測值有延遲。H(s) 也包含這個時間延遲特性。
因此 B(s) = H(s)·C(s)，只有在理想感測器 H(s) = 1 時，B(s) 才等於 C(s)。

Why B(s) ≠ C(s)? Why is H(s) needed?
C(s) is the actual physical quantity (e.g., temperature 80°C), but the controller cannot read temperature directly. The sensor H(s) does two things:
1. Signal Conversion: Converts the physical quantity into an electronic signal (e.g., 4–20 mA, 0–5 V). H(s) includes this conversion ratio (gain).
2. Sensor Dynamics: Sensors have response time (e.g., thermocouple thermal lag), so the measurement is delayed. H(s) captures this dynamic behavior.
Therefore B(s) = H(s)·C(s). Only when H(s) = 1 (ideal sensor) does B(s) equal C(s).

✱ 關鍵概念：閉迴路 (Closed-loop) 控制系統。感測器回饋 B(s) = H(s)·C(s)，與設定值比較產生誤差 E(s) = R(s) − B(s)，驅動控制器調節。

Key Concept: Closed-loop control system. Sensor feedback B(s) = H(s)·C(s) is compared with set point to generate error E(s) = R(s) − B(s), which drives the controller.

信號流程：R(s) → Σ → E(s) → Gc(s) → Gv(s) → Process Gp(s) → C(s) → H(s) → B(s) → Σ（回到比較器）

Signal flow: R(s) → Σ → E(s) → Gc(s) → Gv(s) → Process Gp(s) → C(s) → H(s) → B(s) → Σ (back to comparator)

★ 生活實例：洗澡水溫控制 / Real-Life Example: Shower Temperature Control

想像你在調整蓮蓬頭的水溫，這就是一個完整的閉迴路控制系統：
Imagine adjusting the shower water temperature — this is a complete closed-loop control system:

方塊圖元件 / Component	**對應生活情境 / Real-Life Counterpart**
Set Point R(s)	你心中想要的水溫（如 40°C） Your desired water temperature (e.g., 40°C)
Sensor H(s)	你的手感受到的水溫 Your hand sensing the water temperature
B(s) = H(s)·C(s)	手回傳給大腦的溫度感覺 Temperature sensation fed back to your brain
Comparator (Σ)	大腦比較「現在水溫」vs「想要水溫」 Brain compares current vs desired temperature
E(s) = R−B	誤差：太冷 (E>0) 或太熱 (E<0) Error: too cold (E>0) or too hot (E<0)
Controller Gc(s)	大腦決定旋鈕要轉多少 Brain decides how much to turn the knob
Final Control Gv(s)	手去轉熱水/冷水旋鈕 Hand turning the hot/cold knob
Process Gp(s)	冷熱水混合的物理程序 Physical process of hot & cold water mixing
Output C(s)	實際出水溫度 Actual water temperature at showerhead
Disturbance D(s)	有人沖馬桶 → 水壓驟降 → 水溫突變 Someone flushes toilet → pressure drops → temp change

整個流程不斷循環：感測→比較→決策→動作→再感測，直到水溫穩定。這就是負回饋控制。
The loop runs continuously: sense → compare → decide → act → sense again, until temperature stabilizes. This is negative feedback control.

Unit 3: 化學混合槽與物料平衡

Unit 3: Mixing Tank & Material Balance

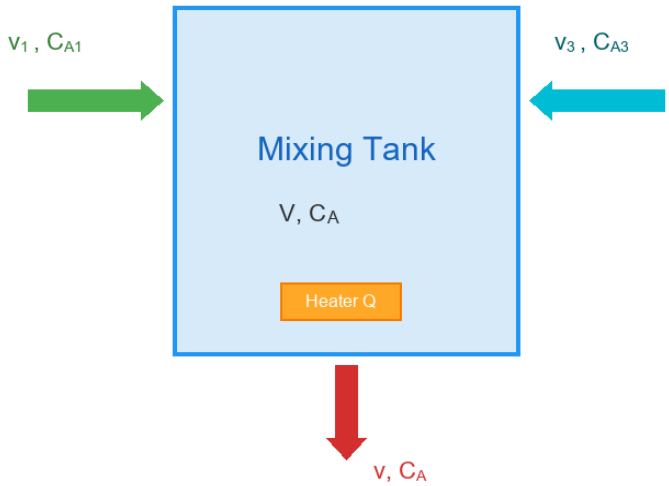
透過化學混合槽系統建立程序動態模型，推導穩態 (Steady State) 與非穩態 (Unsteady State) 物料平衡方程式。

Using a chemical mixing tank system to build a process dynamic model, deriving steady state (SS) and unsteady state (USS) material balance equations.

系統描述 System Description

混合槽有兩個進料流（v_1, C_{A1} 和 v_3, C_{A3}）與一個出料流（v, C_A）。槽內完全混合 (perfectly mixed)，故槽內濃度 = 出口濃度。

The mixing tank has two inlet streams (v_1, C_{A1} and v_3, C_{A3}) and one outlet stream (v, C_A). The tank is perfectly mixed, so the tank concentration equals the outlet concentration.



混合槽系統示意圖 / Mixing Tank System Diagram

變數定義 Variable Definitions

符號 Symbol	意義 Meaning	單位 Unit
C_A	槽內溶液濃度 Tank concentration	mol/L
v, v_1, v_3	體積流量 Volumetric flow rate	L/min
V	槽內液體體積（常數）Tank volume (const.)	L
Q	加熱器輸入熱量 Heater input	kJ/min

T	溫度 Temperature	°C
ρ, C_p	液體密度, 比熱容 Density, specific heat	kg/L, kJ/(kg·°C)

▶ **假設：完全混合 (perfectly mixed) → 槽內濃度 = 出口濃度。$v = v_1 + v_3$，V = 常數。**

Assumptions: Perfectly mixed → tank concentration = outlet concentration. $v = v_1 + v_3$, V = constant.

物料平衡推導 Material Balance Derivation

Step 1 — 基本原理 Fundamental Principle

累積速率 = 輸入速率 − 輸出速率

物質不會憑空產生或消失（質量守恆）。

Accumulation rate = Input rate − Output rate (mass conservation).

Step 2 — 非穩態方程式 Unsteady-State Equation

$$V \cdot dC_A/dt = v_1 \cdot C_{A1} + v_3 \cdot C_{A3} - v \cdot C_A$$

左邊 = 累積項（$V \times dC_A/dt$），右邊 = 淨輸入。V = 常數。

LHS = accumulation ($V \times dC_A/dt$), RHS = net input. V = constant.

Step 3 — 穩態方程式 Steady-State Equation

$$v_1 \cdot C_{A1,s} + v_3 \cdot C_{A3,s} = v \cdot C_{A,s}$$

穩態時累積 = 0，所有變數為常數（下標 s）。

At steady state, accumulation = 0, all variables are constant (subscript s).

Step 4 — USS − SS（減去穩態方程式）

$$V \cdot d(C_A - C_{A,s})/dt = v_3 \cdot (C_{A3} - C_{A3,s}) - v \cdot (C_A - C_{A,s})$$

假設 v_1, C_{A1} 不變（只考慮 C_{A3} 的 step change），消除常數項。

Assuming v_1, C_{A1} unchanged (only step change in C_{A3}), eliminates constant terms.

Step 5 — 引入偏差變數 Deviation Variables

$$Y = C_A - C_{A,s}, \quad X = C_{A3} - C_{A3,s}$$

偏差 = 實際值 − 穩態值。初始條件自動歸零：Y(0) = 0。

Deviation = actual value − steady-state value. Initial conditions automatically zero: Y(0) = 0.

Step 6 — 標準一階形式 Standard First-Order Form

$$\tau \cdot dY/dt + Y = K_p \cdot X$$

$\tau = V/v$ = 時間常數 = 滯留時間 (residence time)，$K_p = v_3/v$ = 穩態增益 (steady-state gain)。

$\tau = V/v$ = time constant = residence time, $K_p = v_3/v$ = steady-state gain.

τ 與 K_p 的物理意義

Physical Meaning of τ and K_p

時間常數 $\tau = V/v$（單位：時間，例如 min）

- 滯留時間（residence time）：流體元素在槽內停留的平均時間。
- 槽體積 V↑ 或總流量 v↓ → τ 變大 → 系統響應*更慢*。
- 在 $t = \tau$ 時，輸出達到最終穩態變化量的 63.2%。
- τ 決定系統對輸入變化的*反應速度*。

穩態增益 $K_p = v_3/v$（無因次）

- 系統達到新穩態後，輸出變化與輸入變化的比值：$\Delta y_{ss} = K_p \cdot A$。
- $K_p > 1$：輸出被*放大*；$K_p < 1$：輸出被*衰減*；$K_p = 1$：輸出等於輸入。
- K_p 決定輸出對給定輸入變化的*變化幅度*。

Time constant $\tau = V/v$ (unit: time, e.g. min)

- Residence time: the average time a fluid element stays inside the tank.
- Larger V ↑ or slower v ↓ → larger τ → *slower* system response.
- At $t = \tau$, the output reaches 63.2% of its final steady-state change.
- τ determines *how fast* the system responds to an input change.

Steady-state gain $K_p = v_3/v$ (dimensionless)

- The ratio of output change to input change at the new steady state: $\Delta y_{ss} = K_p \cdot A$.
- $K_p > 1$: output is *amplified*; $K_p < 1$: output is *attenuated*; $K_p = 1$: output equals input.
- K_p determines *how much* the output changes in response to a given input change.

⚠ 為什麼用偏差變數？

1. 初始條件自動歸零：Y(0) = 0

2. Laplace 轉換更簡潔

3. 方程式更精簡（消除常數項）

Why use deviation variables?

1. Initial conditions automatically zero: Y(0) = 0

2. Laplace transform becomes simpler

3. Equations are more concise (constant terms eliminated)

Unit 4: 一階系統求解與 Step Response

Unit 4: First-Order System Solution & Step Response

用積分因子法 (Integrating Factor Method) 求解一階 ODE，並觀察時間常數 τ 與穩態增益 K_p 的影響。

Solving the first-order ODE using the Integrating Factor Method, and observing the effects of time constant τ and steady-state gain K_p.

▶ **Step Input（階躍輸入）**：首先定義單位階躍函數（unit step function）：

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

大小為 A 的階躍輸入可表示為：

$$X(t) = Au(t) = \begin{cases} 0, & t < 0 \\ A, & t \geq 0 \end{cases}$$

這是控制工程中最基本、最常用的「標準測試信號」。

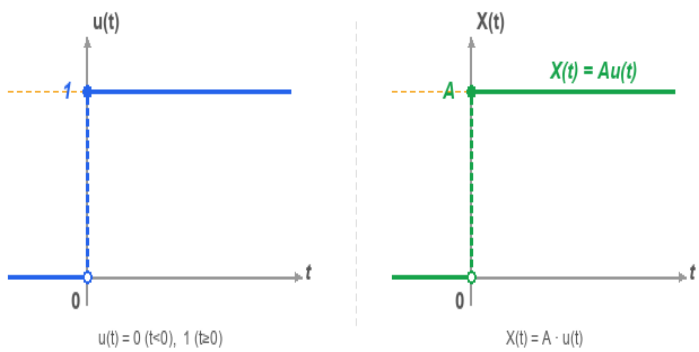
Step Input: First, define the unit step function:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

A step input of magnitude A is then expressed as:

$$X(t) = Au(t) = \begin{cases} 0, & t < 0 \\ A, & t \geq 0 \end{cases}$$

This is the most fundamental and widely used **standard test signal** in control engineering.



Step Input 示意圖 / Step Input Illustration

▶ **Step Response（階躍響應）**：系統對 Step Input 產生的輸出 Y(t)。它揭示了系統的兩個核心特性： • 時間常數 τ：系統多快達到新的穩態？（5τ ≈ 99%） • 穩態增益 K_p：輸入變 A，輸出最終變多少？（$K_p \cdot A$）

Step Response: *The system output Y(t) resulting from a Step Input. It reveals two core system characteristics: • Time constant τ: How fast does the system reach the new steady state? (5τ ≈ 99%) • Steady-state gain K_p: If input changes by A, how much does the output ultimately change? ($K_p \cdot A$)*

✱ 為什麼用 Step Input 測試？ 1. 現實中常見：閥門開關、設定值調整、進料切換都近似 step change 2. 數學上好解：常數輸入讓 ODE 有解析解 3. 一次揭示 τ 和 K_p：從一條響應曲線即可辨識系統參數

Why use Step Input for testing? *1. Common in practice: valve switching, setpoint changes, feed transitions ≈ step changes 2. Mathematically tractable: constant input yields an analytical ODE solution 3. Reveals τ and K_p at once: system parameters identified from a single response curve*

積分因子法逐步求解 Step-by-Step Solution

Step 1 — 一階 ODE

$$\tau \cdot dY/dt + Y = K_p \cdot X(t)$$

大小為 A 的階躍輸入：X(t) = Au(t)，即 t ≥ 0 時 X(t) = A。

Step input of magnitude A: X(t) = Au(t), i.e., X(t) = A for t ≥ 0.

Step 2 — 整理 Rearrange

$$dY/dt + (1/\tau)\cdot Y = K_p\cdot A/\tau$$

Step 3 — 積分因子 Integrating Factor

$$\mu = \exp(t/\tau)$$

Step 4 — 乘以積分因子 Multiply by Integrating Factor

$$d/dt\,[\exp(t/\tau) \cdot Y] = (K_p\cdot A/\tau) \cdot \exp(t/\tau)$$

Step 5 — 積分 + 初始條件 Integrate with Y(0) = 0

$$\exp(t/\tau) \cdot Y = K_p\cdot A\cdot\exp(t/\tau) + C_1 \;\Rightarrow\; C_1 = -K_p\cdot A$$

代入初始條件 Y(0) = 0。

Applying initial condition Y(0) = 0.

Step 6 — 最終解 Final Solution (Unit Step Response)

$$Y(t) = K_p\cdot A\cdot(1 - \exp(-t/\tau))$$

$t \to \infty$ 時 $Y \to K_p\cdot A$（穩態值）。

As $t \to \infty$, $Y \to K_p\cdot A$ (steady-state value).

✱ 核心對應關係：輸入 x 的階躍輸入 Au(t) → 輸出 y 獲得增益 $K_p\cdot A$ $\Delta x = A \Rightarrow \Delta y_{ss}$ ($_{ss}$ = steady-state 穩態值) = $K_p \cdot A$

Core relationship: A step input of magnitude A A in input x → output y gains $K_p \cdot A$ $\Delta x = A$ ⇒ Δy_{ss} (y_{ss} = steady-state value, 穩態值) = $K_p \cdot A$

Step Response 特性 Characteristics

時間 Time	$Y(t) / (K_p \cdot A)$	達到比例 % of Final
$t = \tau$	$1 - e^{-1} = 0.632$	63.2%
$t = 2\tau$	$1 - e^{-2} = 0.865$	86.5%
$t = 3\tau$	$1 - e^{-3} = 0.950$	95.0%
$t = 4\tau$	$1 - e^{-4} = 0.982$	98.2%
$t = 5\tau$	$1 - e^{-5} = 0.993$	99.3%

✱ 重要觀察： • τ↑ → 響應更慢 • $K_p \cdot A$ = 最終穩態值 • 由 $Y(t) = Kp \cdot A \cdot (1 - e^{(-t/\tau)})$，到達穩態值比例為 $1 - e^{(-t/\tau)}$： $t = \tau$：$1 - e^{(-1)} \approx 63.2\%$，$t = 3\tau$：$1 - e^{(-3)} \approx 95.0\%$，$t = 5\tau$：$1 - e^{(-5)} \approx 99.3\%$

Key observations: • τ↑ → slower response • $K_p \cdot A$ = final steady-state value • From $Y(t) = Kp \cdot A \cdot (1 - e^{(-t/\tau)})$, fraction = $1 - e^{(-t/\tau)}$: $t = \tau$: $1-e^{(-1)} \approx 63.2\%$, $t = 3\tau$: $1-e^{(-3)} \approx 95.0\%$, $t = 5\tau$: $1-e^{(-5)} \approx 99.3\%$

Unit 5: 能量平衡與標準一階系統

Unit 5: Energy Balance & Standard First-Order System

能量平衡推導的結構與物料平衡完全相同，最終都歸結為標準一階系統形式。

The energy balance derivation has the same structure as the material balance, both reducing to the standard first-order system form.

能量平衡推導 Energy Balance Derivation

假設：ρ (密度)、C_p (比熱容) 為常數。

Assumptions: ρ (density) and C_p (specific heat) are constant.

Step 1 — 穩態 Steady State (SS)

$$\rho \cdot C_p \cdot v \cdot T_{i,s} + Q_s = \rho \cdot C_p \cdot v \cdot T_s$$

Step 2 — 非穩態 Unsteady State (USS)

$$\rho \cdot C_p \cdot V \cdot dT/dt = \rho \cdot C_p \cdot v \cdot T_i + Q - \rho \cdot C_p \cdot v \cdot T$$

Step 3 — USS − SS（設 Q 不變, Step Change in T_i）

$$\rho \cdot C_p \cdot V \cdot d(T-T_s)/dt = \rho \cdot C_p \cdot v \cdot (T_i - T_{i,s}) - \rho \cdot C_p \cdot v \cdot (T - T_s)$$

假設 Q 不變（no control），只有 T_i 有 step change。

Assuming Q unchanged (no control), only T_i has a step change.

Step 4 — 偏差變數 & 標準形式

$$Y = T - T_s,\ \ X = T_i - T_{i,s} \quad (V/v) \cdot dY/dt + Y = 1 \cdot X$$

$\rho \cdot C_p$ 同時約分！$\tau = V/v$，$K_p = 1$。與物料平衡形式完全相同！

$\rho \cdot C_p$ cancels out! $\tau = V/v$, $K_p = 1$. Same form as material balance!

物料 vs 能量 完整對照表

Material vs Energy Complete Comparison Table

項目 Item	物料平衡 Material	能量平衡 Energy
守恆量 Conserved	質量 (C_A) Mass	能量；T 為狀態變數 Energy
SS 方程式	$v_1C_{A1,s} + v_3C_{A3,s} = vC_{A,s}$	$\rho C_p \cdot v \cdot T_{i,s} + Q_s = \rho C_p \cdot v \cdot T_s$
USS 方程式	$V \cdot dC_A/dt = v_1C_{A1} + v_3C_{A3} - vC_A$	$\rho C_p V \cdot dT/dt = \rho Cpv \cdot T_i + Q - \rho Cpv \cdot T$
偏差變數 Deviation	$Y = C_A - C_{A,s}$	$Y = T - T_s$
標準形式 Standard	$\tau \cdot dY/dt + Y = K_p \cdot X$	$\tau \cdot dY/dt + Y = K_p \cdot X$
τ	V / v	V / v
K_p	v_3 / v	1

Step Response	$Y = K_p \cdot A \cdot (1-e^{(-t/\tau)})$	$Y = A \cdot (1-e^{(-t/\tau)})$

✱ 核心結論：任何滿足 $\tau \cdot dY/dt + Y = K_p \cdot X$ 的系統都是「一階系統」。

「x 的階躍輸入 Au(t) → y 獲得增益 K_p」是一階系統最重要的輸入-輸出關係。

Core conclusion: Any system satisfying $\tau \cdot dY/dt + Y = K_p \cdot X$ is a 'first-order system'. 'Step input Au(t) in x → y gains K_p' is the most important I/O relationship of first-order systems.

Unit 6: Laplace 轉換基礎

Unit 6: Laplace Transform Fundamentals

Laplace 轉換是分析線性系統的核心工具，將時域 ODE 轉為 s 域代數方程式。

The Laplace transform is the core tool for analyzing linear systems, converting time-domain ODEs into s-domain algebraic equations.

▶ **為什麼要從時域轉到 s 域？ 1. 微分方程式 → 代數方程式：求解從「解 ODE」簡化為「多項式運算」 2. 轉移函數 G(s)：直接顯示系統增益 K_p 與時間常數 τ 3. 方塊圖運算：串聯=相乘、並聯=相加、回饋=閉迴路公式 4. 穩定性判斷：從極點（分母根）位置即可判定系統穩定性**

Why transform from time domain to s-domain? 1. Differential equations → Algebraic equations: solving simplifies from 'solve ODE' to 'polynomial algebra' 2. Transfer function G(s): directly reveals system gain K_p and time constant τ 3. Block diagram algebra: series=multiply, parallel=add, feedback=closed-loop formula 4. Stability: determined by pole (denominator root) locations

Table 2.1 — Laplace 轉換對照表

Common Laplace Transform Table

f(t) (t≥0)	F(s) = L{f}	名稱 Name
u(t)	1/s	單位階躍 Unit Step
t u(t)	$1/s^2$	斜坡 Ramp
t^n u(t)	$n!/s^{(n+1)}$	冪次 Power
e^{-at} u(t)	1/(s+a)	指數衰減 Exponential Decay
$t^n e^{-at}$ u(t)	$n!/(s+a)^{(n+1)}$	平移冪次 Shifted Power
sin(kt) u(t)	$k/(s^2+k^2)$	正弦 Sine
cos(kt) u(t)	$s/(s^2+k^2)$	餘弦 Cosine
sinh(kt) u(t)	$k/(s^2-k^2)$	雙曲正弦 Hyperbolic Sine
cosh(kt) u(t)	$s/(s^2-k^2)$	雙曲餘弦 Hyperbolic Cosine
e^{-at} sin(kt) u(t)	$k/[(s+a)^2+k^2]$	衰減正弦 Damped Sine
e^{-at} cos(kt) u(t)	$(s+a)/[(s+a)^2+k^2]$	衰減餘弦 Damped Cosine
δ(t)	1	單位脈衝 Unit Impulse

Table 2.1 完整推導 Complete Derivations

以下為 Table 2.1 中每個拉普拉斯轉換對的完整推導過程。

Below are the complete derivations for each Laplace transform pair in Table 2.1.

註：Re(s) 表示複數 s = σ + jω 的實部 σ，即 Re(s) = σ。收斂條件 Re(s) > 0 確保積分 e^{-st} 在 t→∞ 時趨近於零。

Note: Re(s) denotes the real part of the complex variable s = σ + jω, i.e., Re(s) = σ. The convergence condition Re(s) > 0 ensures the integrand $e^{-st} \to 0$ as $t \to \infty$.

1. 單位階躍函數 Unit Step Function: u(t)

轉換 (Transform): L{u(t)} = 1/s

步驟 1 (Step 1): 由定義 (By definition):

$F(s) = \int_0^\infty u(t)\, e^{(-st)}\, dt$

因為 t > 0 時 u(t) = 1 (Since u(t)=1 for t>0):

$F(s) = \int_0^\infty e^{(-st)}\, dt$

步驟 2 (Step 2): 積分 (Integrate):

$F(s) = [e^{(-st)} / (-s)]_0^\infty$

$= \lim(t\to\infty)\, e^{(-st)}/(-s) - e^0/(-s)$

步驟 3 (Step 3): 當 Re(s) > 0 時，$e^{(-st)} \to 0$ (When Re(s)>0, $e^{(-st)}\to 0$):

$F(s) = 0 - 1/(-s) = 1/s$

結果 (Result): L{u(t)} = 1/s, Re(s) > 0

應用 (Application): 階躍輸入是控制系統中最基本的測試信號。(The step input is the most fundamental test signal in control systems.)

---

2. 斜坡函數 Ramp Function: t·u(t)

轉換 (Transform): L{t·u(t)} = 1/s²

步驟 1 (Step 1): 由定義 (By definition):

$F(s) = \int_0^\infty t \cdot e^{(-st)}\, dt$

步驟 2 (Step 2): 使用分部積分，令 u=t, dv=$e^{(-st)}$dt (Integration by parts, let u=t, dv=$e^{(-st)}$dt):

$F(s) = [t \cdot e^{(-st)}/(-s)]_0^\infty - \int_0^\infty e^{(-st)}/(-s)\, dt$

$= 0 + (1/s) \int_0^\infty e^{(-st)}\, dt$

步驟 3 (Step 3): 利用已知結果 $\int_0^\infty e^{(-st)}\, dt = 1/s$ (Using known result):

$F(s) = (1/s)(1/s) = 1/s^2$

結果 (Result): L{t·u(t)} = 1/s², Re(s) > 0

應用 (Application): 斜坡輸入用於測試系統的穩態誤差。(Ramp input is used to test system steady-state error.)

---

3. 冪函數 Power Function: t^n·u(t)

轉換 (Transform): L{t^n·u(t)} = n!/$s^{(n+1)}$

步驟 1 (Step 1): 由定義 (By definition):

$F(s) = \int_0^\infty t^n \cdot e^{(-st)}\, dt$

步驟 2 (Step 2): 換元 令 $x = st$，則 $t = x/s$，$dt = dx/s$ (Substitution: let x=st):

$F(s) = \int_0^\infty (x/s)^n \cdot e^{(-x)} \cdot (dx/s)$

$= (1/s^{(n+1)}) \int_0^\infty x^n \cdot e^{(-x)}\, dx$

步驟 3 (Step 3): 識別 Gamma 函數 $\Gamma(n+1) = \int_0^\infty x^n e^{(-x)}\, dx = n!$ (Recognize Gamma function):

$F(s) = n! / s^{(n+1)}$

結果 (Result): $L\{t^n \cdot u(t)\} = n!/s^{(n+1)}$, $n = 0,1,2,...$

應用 (Application): 當 n=1 即為斜坡函數，n=2 為拋物線函數。(When n=1 this is the ramp; n=2 gives the parabolic function.)

---

4. 指數函數 Exponential: $e^{(-at)} \cdot u(t)$

轉換 (Transform): $L\{e^{(-at)}\} = 1/(s+a)$

步驟 1 (Step 1): 由定義 (By definition):

$F(s) = \int_0^\infty e^{(-at)} \cdot e^{(-st)}\, dt = \int_0^\infty e^{(-(s+a)t)}\, dt$

步驟 2 (Step 2): 積分 (Integrate):

$F(s) = [e^{(-(s+a)t)} / (-(s+a))]_0^\infty$

步驟 3 (Step 3): 當 $Re(s+a) > 0$ 時 (When $Re(s+a)>0$):

$F(s) = 0 - 1/(-(s+a)) = 1/(s+a)$

結果 (Result): $L\{e^{(-at)}\} = 1/(s+a)$, $Re(s) > -a$

應用 (Application): 指數衰減是一階系統的自然響應。(Exponential decay is the natural response of first-order systems.)

---

5. 冪指數函數 Power-Exponential: $t^n e^{(-at)} \cdot u(t)$

轉換 (Transform): $L\{t^n e^{(-at)}\} = n!/(s+a)^{(n+1)}$

步驟 1 (Step 1): 利用頻移性質 (Using frequency-shift property):

若 $L\{f(t)\} = F(s)$，則 $L\{e^{(-at)}f(t)\} = F(s+a)$

(If $L\{f(t)\}=F(s)$, then $L\{e^{(-at)}f(t)\}=F(s+a)$)

步驟 2 (Step 2): 已知 $L\{t^n\} = n!/s^{(n+1)}$ (Known: $L\{t^n\}=n!/s^{(n+1)}$):

$L\{t^n \cdot e^{(-at)}\} = n!/(s+a)^{(n+1)}$

步驟 3 (Step 3): 或者直接計算 (Or compute directly):

$\int_0^\infty t^n e^{(-at)} e^{(-st)}\, dt = \int_0^\infty t^n e^{(-(s+a)t)}\, dt$

換元 $x=(s+a)t$ 即得 $n!/(s+a)^{(n+1)}$ (Substituting $x=(s+a)t$ gives the result)

結果 (Result): $L\{t^n e^{(-at)}\} = n!/(s+a)^{(n+1)}$

應用 (Application): 用於部分分式展開中的重複極點。(Used for repeated poles in partial fraction expansion.)

---

6. 正弦函數 Sine: sin(kt)·u(t)

轉換 (Transform): $L\{\sin(kt)\} = k/(s^2+k^2)$

步驟 1 (Step 1): 利用歐拉公式 (Using Euler's formula):

$\sin(kt) = (e^{(jkt)} - e^{(-jkt)}) / (2j)$

步驟 2 (Step 2): 取拉普拉斯轉換 (Take Laplace transform):

$L\{\sin(kt)\} = (1/(2j)) [1/(s-jk) - 1/(s+jk)]$

步驟 3 (Step 3): 合併分式 (Combine fractions):

$= (1/(2j)) \cdot [(s+jk)-(s-jk)] / [(s-jk)(s+jk)]$
$= (1/(2j)) \cdot (2jk) / (s^2+k^2)$
$= k/(s^2+k^2)$

結果 (Result): $L\{\sin(kt)\} = k/(s^2+k^2)$, $Re(s) > 0$

應用 (Application): 正弦信號用於頻率響應分析。(Sinusoidal signals are used in frequency response analysis.)

---

7. 餘弦函數 Cosine: cos(kt)·u(t)

轉換 (Transform): $L\{\cos(kt)\} = s/(s^2+k^2)$

步驟 1 (Step 1): 利用歐拉公式 (Using Euler's formula):

$\cos(kt) = (e^{(jkt)} + e^{(-jkt)}) / 2$

步驟 2 (Step 2): 取拉普拉斯轉換 (Take Laplace transform):

$L\{\cos(kt)\} = (1/2) [1/(s-jk) + 1/(s+jk)]$

步驟 3 (Step 3): 合併分式 (Combine fractions):

$= (1/2) \cdot [(s+jk)+(s-jk)] / [(s-jk)(s+jk)]$
$= (1/2) \cdot (2s) / (s^2+k^2)$
$= s/(s^2+k^2)$

結果 (Result): $L\{\cos(kt)\} = s/(s^2+k^2)$, $Re(s) > 0$

應用 (Application): 餘弦函數常見於振盪系統響應。(Cosine appears in oscillatory system responses.)

---

8. 雙曲正弦 Hyperbolic Sine: sinh(kt)·u(t)

轉換 (Transform): $L\{\sinh(kt)\} = k/(s^2-k^2)$

步驟 1 (Step 1): 由定義 (By definition):

$\sinh(kt) = (e^{(kt)} - e^{(-kt)}) / 2$

步驟 2 (Step 2): 取拉普拉斯轉換 (Take Laplace transform):

$L\{\sinh(kt)\} = (1/2) [1/(s-k) - 1/(s+k)]$

步驟 3 (Step 3): 合併分式 (Combine fractions):

$= (1/2) \cdot [(s+k)-(s-k)] / [(s-k)(s+k)]$

$= (1/2) \cdot (2k) / (s^2-k^2)$

$= k/(s^2-k^2)$

結果 (Result): $L\{\sinh(kt)\} = k/(s^2-k^2)$, $Re(s) > |k|$

應用 (Application): 雙曲函數出現在不穩定系統的響應中。(Hyperbolic functions appear in unstable system responses.)

---

9. 雙曲餘弦 Hyperbolic Cosine: $\cosh(kt)\cdot u(t)$

轉換 (Transform): $L\{\cosh(kt)\} = s/(s^2-k^2)$

步驟 1 (Step 1): 由定義 (By definition):

$\cosh(kt) = (e^{(kt)} + e^{(-kt)}) / 2$

步驟 2 (Step 2): 取拉普拉斯轉換 (Take Laplace transform):

$L\{\cosh(kt)\} = (1/2) [1/(s-k) + 1/(s+k)]$

步驟 3 (Step 3): 合併分式 (Combine fractions):

$= (1/2) \cdot [(s+k)+(s-k)] / [(s-k)(s+k)]$

$= (1/2) \cdot (2s) / (s^2-k^2)$

$= s/(s^2-k^2)$

結果 (Result): $L\{\cosh(kt)\} = s/(s^2-k^2)$, $Re(s) > |k|$

應用 (Application): cosh(kt) 與 sinh(kt) 可透過指數函數的線性組合得出。(cosh and sinh can be derived as linear combinations of exponentials.)

---

10. 衰減正弦 Damped Sine: $e^{(-at)}\sin(kt)\cdot u(t)$

轉換 (Transform): $L\{e^{(-at)}\sin(kt)\} = k/[(s+a)^2+k^2]$

步驟 1 (Step 1): 利用頻移性質 (Using frequency-shift property):

若 $L\{f(t)\} = F(s)$，則 $L\{e^{(-at)}f(t)\} = F(s+a)$

(If $L\{f(t)\}=F(s)$, then $L\{e^{(-at)}f(t)\}=F(s+a)$)

步驟 2 (Step 2): 已知 $L\{\sin(kt)\} = k/(s^2+k^2)$ (Known result):

$L\{e^{(-at)}\sin(kt)\} = k/((s+a)^2+k^2)$

步驟 3 (Step 3): 展開驗證 (Verify by expanding):

$(s+a)^2+k^2 = s^2+2as+a^2+k^2$

故 $L\{e^{(-at)}\sin(kt)\} = k/(s^2+2as+a^2+k^2)$

結果 (Result): $L\{e^{(-at)}\sin(kt)\} = k/[(s+a)^2+k^2]$

應用 (Application): 衰減正弦是欠阻尼二階系統的典型響應。(Damped sine is the typical response of underdamped second-order systems.)

11. 衰減餘弦 Damped Cosine: $e^{(-at)}\cos(kt)\cdot u(t)$

轉換 (Transform): $L\{e^{(-at)}\cos(kt)\} = (s+a)/[(s+a)^2+k^2]$

步驟 1 (Step 1): 利用頻移性質 (Using frequency-shift property):

若 $L\{f(t)\} = F(s)$，則 $L\{e^{(-at)}f(t)\} = F(s+a)$

步驟 2 (Step 2): 已知 $L\{\cos(kt)\} = s/(s^2+k^2)$ (Known result):

$L\{e^{(-at)}\cos(kt)\} = (s+a)/((s+a)^2+k^2)$

步驟 3 (Step 3): 這是頻移性質的直接應用 (Direct application of frequency-shift):

將 $F(s) = s/(s^2+k^2)$ 中的 s 替換為 (s+a) (Replace s with (s+a) in F(s))

結果 (Result): $L\{e^{(-at)}\cos(kt)\} = (s+a)/[(s+a)^2+k^2]$

應用 (Application): 衰減餘弦與衰減正弦共同描述欠阻尼系統的完整響應。(Damped cosine and damped sine together describe the complete underdamped response.)

---

12. 單位脈衝函數 Unit Impulse: δ(t)

轉換 (Transform): $L\{\delta(t)\} = 1$

步驟 1 (Step 1): Dirac delta 函數的篩選性質 (Sifting property of Dirac delta):

$\int_{-\infty}^{\infty} f(t)\cdot\delta(t-t_0)\,dt = f(t_0)$

步驟 2 (Step 2): 將此性質應用於拉普拉斯轉換 (Apply to Laplace transform):

$L\{\delta(t)\} = \int_0^{\infty} \delta(t)\cdot e^{(-st)}\,dt$

由篩選性質，在 t=0 處取值 (By sifting property, evaluate at t=0):

$= e^{(-s\cdot 0)} = e^0 = 1$

步驟 3 (Step 3): 注意 δ(t) 是 u(t) 的導數 (Note: δ(t) is the derivative of u(t)):

$L\{u'(t)\} = s\cdot L\{u(t)\} - u(0^-) = s\cdot(1/s) - 0 = 1$

此結果與直接計算一致 (This is consistent with the direct calculation)

結果 (Result): $L\{\delta(t)\} = 1$

應用 (Application): 脈衝響應是系統的基本特徵，等於轉移函數的反拉普拉斯轉換。(The impulse response is the fundamental system characteristic, equal to the inverse Laplace transform of the transfer function.)

重要性質 Key Properties

性質 Property	時域 f(t)	s 域 F(s)
導數 Derivative	df/dt	$s\cdot F(s) - f(0)$
積分 Integral	$\int_0^t f(\xi)d\xi$	F(s)/s
s 軸位移 s-Shifting	$e^{(-at)}\cdot f(t)$	F(s+a)
t 軸位移 t-Shifting	$f(t-a)\cdot u(t-a)$	$e^{(-as)}\cdot F(s)$

Final Value Theorem	$\lim(t\to\infty)\ f(t)$	$\lim(s\to 0)\ s\cdot F(s)$
Initial Value Theorem	$\lim(t\to 0^{+})\ f(t)$	$\lim(s\to\infty)\ s\cdot F(s)$

▶ s 軸位移的應用：$e^{(-at)}\cdot\sin(\omega t) \leftrightarrow \omega/[(s+a)^2+\omega^2]$ — 在 Example 3.4（複數根）中使用。

Application of s-shifting: $e^{(-at)}\cdot\sin(\omega t) \leftrightarrow \omega/[(s+a)^2+\omega^2]$ — used in Example 3.4 (complex roots).

Unit 7: 轉移函數 (Transfer Function)

Unit 7: Transfer Function

將一階系統從時域轉到 s 域，建立輸入與輸出之間的代數關係。

Transform the first-order system from time domain to s-domain, establishing the algebraic relationship between input and output.

推導 Derivation

Step 1 — 時域方程式 Time-Domain Equation

$$\tau \cdot dY/dt + Y = K_p \cdot X$$

Step 2 — 取 Laplace 轉換 Take Laplace Transform (Y(0)=0)

$$\tau \cdot s \cdot Y(s) + Y(s) = K_p \cdot X(s)$$

偏差變數初始條件 Y(0)=0 使導數項簡化。

Deviation variable initial condition Y(0)=0 simplifies the derivative term.

Step 3 — 提出 Y(s) Factor out Y(s)

$$Y(s) \cdot (\tau s + 1) = K_p \cdot X(s)$$

Step 4 — 轉移函數 Transfer Function

$$G(s) = Y(s)/X(s) = K_p / (\tau s + 1)$$

這就是一階系統轉移函數。分母根 $s = -1/\tau$ 是極點 (Pole)。

This is the first-order transfer function. The denominator root $s = -1/\tau$ is the Pole.

✱ Laplace 轉換的偏差變數對應：

Y(s) = y 偏差的 Laplace 轉換

X(s) = x 偏差的 Laplace 轉換

轉移函數 = 輸出/輸入（在 s 域）

Deviation variable correspondence:

Y(s) = Laplace transform of y deviation

X(s) = Laplace transform of x deviation

Transfer function = Output/Input (in s-domain)

⚠ 極點 (Pole) 的物理意義 / Physical Meaning of Poles

極點 = 轉移函數分母 = 0 的根，它直接決定系統被擾動後的時間行為：

A Pole is the root where the transfer function denominator = 0. It directly determines how the system behaves after being disturbed:

極點位置 / Pole Location	**時域行為 / Time-Domain**	**物理意義 / Physical Meaning**
實數負極點 $s = -1/\tau$	$e^{(-t/\tau)}$ 指數衰減	穩定，τ 越小衰減越快 Stable; smaller τ → faster decay
s = 0（原點 / origin）	常數 / 積分行為 constant / integrating	系統無自然回穩能力 No natural return to steady state
實數正極點 s = +a	$e^{(+at)}$ 指數成長	不穩定，輸出無限增大 Unstable; output grows unbounded
複數極點 s = −a ± bi	$e^{(-at)}\sin(bt)$ 振盪衰減	a 控制衰減速率，b 控制振盪頻率 a: decay rate, b: oscillation freq

一階系統的直接連結：$G(s) = Kp/(\tau s+1)$ 的極點在 $s = -1/\tau$，極點位置就是時間常數的倒數。

Direct link: $G(s) = Kp/(\tau s+1)$ has its pole at $s = -1/\tau$ — the pole location is the reciprocal of the time constant.

一句話總結：極點告訴你「系統被擾動後，會以什麼方式回到（或離開）穩態」。

In one sentence: Poles tell you how the system returns to (or departs from) steady state after being disturbed.

Final Value Theorem 驗證 Verification

對 step input $X(s) = A/s$ 施加 Final Value Theorem：

Applying Final Value Theorem to step input $X(s) = A/s$:

$$\lim(s\to 0)\ s \cdot G(s) \cdot (A/s) = \lim(s\to 0)\ K_p \cdot A/(\tau s+1) = K_p \cdot A$$

與時域解 $Y(\infty) = K_p \cdot A$ 完全一致！

Consistent with time-domain solution $Y(\infty) = K_p \cdot A$!

Unit 8: 部分分式展開 & 根的類型

Unit 8: Partial Fraction Expansion & Root Types

反 Laplace 轉換的核心技巧。五種根的類型決定了系統的時域行為。

The core technique for inverse Laplace transform. Five types of roots determine the system's time-domain behavior.

類型 1：實數相異根 Distinct Real Roots

▶ **Example 2.3 / 3.2：F(s) = 1/[s(5s+1)]　根 = 0, −1/5　→ SS gain = 1, Time constant = 5**

Roots = 0, −1/5 → Steady-state gain = 1, Time constant = 5

Step 1 — 設定 Setup

$$1/[s(5s+1)] = A/s + B/(5s+1)$$

Step 2 — 求 A（令 s=0）

$$A = [1/(5s+1)]|_{(s=0)} = 1$$

Cover-up method：遮住 s，代入 s=0。

Cover-up method: cover s, substitute s=0.

Step 3 — 求 B（令 s=−1/5）

$$B = [1/s]|_{(s=-1/5)} = -5$$

遮住 (5s+1)，代入 s=−1/5。

Cover (5s+1), substitute s=−1/5.

Step 4 — 反轉換 Inverse Transform

$$f(t) = 1 - e^{(-t/5)} = K_p \cdot A \cdot (1 - e^{(-t/\tau)})$$

與 step response 完全吻合！

Matches the step response exactly!

類型 2：重根 Repeated Roots

▶ **Example：F(s) = N(s)/[s(s+1)²(s+2)]　根 = 0, −1, −1, −2（−1 為重根）**

Roots = 0, −1, −1, −2 (−1 is a repeated root)

$$F(s) = A/s + B_1/(s+1) + B_2/(s+1)^2 + C/(s+2)$$

⚠ 關鍵：重根 $(s+1)^2$ 必須拆成兩項。B_2 用 cover-up 法，B_1 需微分。 反轉換：$L^{-1}\{1/(s+a)^2\} = t \cdot e^{(-at)}$

Key: Repeated root $(s+1)^2$ must be split into two terms. B_2 uses cover-up, B_1 requires differentiation.
Inverse: $L^{-1}\{1/(s+a)^2\} = t\cdot e^{(-at)}$

類型 3：共軛複數根 Complex Conjugate Roots

▶ **Ex. 3.4：$F(s) = N(s)/[s((s+1)^2+1)]$　根 = 0, −1±i（共軛複數根）**

Roots = 0, −1±i (complex conjugate roots)

$$F(s) = A/s + (Bs+C)/(s^2+2s+2)$$

反轉換含振盪項：$f(t) = A + e^{(-t)}(D\cdot\cos t + E\cdot\sin t)$

Inverse contains oscillatory terms: $f(t) = A + e^{(-t)}(D\cdot\cos t + E\cdot\sin t)$

✱ **使用 s 軸位移：$e^{(-at)}\cdot\sin(\omega t) \leftrightarrow \omega/[(s+a)^2+\omega^2]$**

Using s-axis shifting: $e^{(-at)}\cdot\sin(\omega t) \leftrightarrow \omega/[(s+a)^2+\omega^2]$

類型 4：正實數根 Positive Real Roots

▶ **Example：$F(s) = N(s)/[s(s+1)(s-1)]$　根 = 0, −1, +1（含正實數根）**

Roots = 0, −1, +1 (contains a positive real root)

⚠ **不穩定系統！正根 s = +1 → $e^{(+t)}$ 指數增長（發散）。系統 Unstable（Ch.13 討論）。**

Unstable system! Positive root s = +1 → $e^{(+t)}$ exponential growth (diverges). System is Unstable (discussed in Ch.13).

類型 5：純虛數根 Pure Imaginary Roots

▶ **Example：$F(s) = N(s)/[(s+1)(s^2+4)]$　根 = −1, ±2i（純虛數根）**

Roots = −1, ±2i (pure imaginary roots)

純虛數根 ±ωi（此處 ω=2）：

- 產生等幅振盪 (sustained oscillation)，永不衰減
- sin 2t 和 cos 2t 項，振幅不隨時間改變
- 系統處於臨界穩定 (marginally stable) 的邊界

Pure imaginary roots ±ωi (here ω=2):

- *Produce sustained oscillation, never decaying*
- *sin 2t and cos 2t terms with constant amplitude*
- *System is at the boundary of marginal stability*

$$f(t) = A\cdot e^{(-t)} + B\cdot\cos 2t + C\cdot\sin 2t$$

五種根的行為總結 Summary of 5 Root Types

根的類型 Root Type	s 域形式 Example	時域行為 Time-Domain	穩定性 Stability
實數相異根 Distinct Real	0, −1/5	$K_p \cdot A \cdot (1-e^{(-t/\tau)})$	穩定 Stable
重根 Repeated	−1, −1	$t \cdot e^{(-t)}$	穩定 Stable
共軛複數根 Complex Conj.	−1±i	$e^{(-t)}(\cos t + \sin t)$	穩定（衰減振盪） Stable (damped)
正實數根 Positive Real	+1	$e^{(+t)}$ 發散	不穩定 Unstable
純虛數根 Pure Imaginary	±2i	$\cos 2t + \sin 2t$	臨界穩定 Marginally Stable

總結 Summary

恭喜完成 Chapter 1 & 2！你已學習：

Congratulations on completing Chapter 1 & 2! You have learned:

- ✓ 程序控制的五大原因：安全、環保、品質、效率、獲利

  Five reasons for process control: safety, environment, quality, efficiency, profitability

- ✓ 閉迴路控制系統方塊圖的各元件功能

  Functions of each component in the closed-loop control system block diagram

- ✓ 物料平衡建模：穩態 → 非穩態 → 偏差變數 → 一階標準形式

  Material balance modeling: SS → USS → deviation variables → first-order standard form

- ✓ 積分因子法求解一階 ODE → Step Response: $Y(t) = K_p \cdot A \cdot (1 - e^{(-t/\tau)})$

  Integrating factor method for first-order ODE → Step Response: $Y(t) = K_p \cdot A \cdot (1 - e^{(-t/\tau)})$

- ✓ 能量平衡與物料平衡的結構一致性

  Structural consistency between energy and material balance

- ✓ Laplace 轉換、s/t 軸位移、Final/Initial Value Theorem

  Laplace transform, s/t-axis shifting, Final/Initial Value Theorem

- ✓ 轉移函數 $G(s) = K_p/(\tau s+1)$ 及其極點意義

  Transfer function $G(s) = K_p/(\tau s+1)$ *and its pole significance*

- ✓ 部分分式展開的 5 種根的類型與穩定性判斷

  5 root types in partial fraction expansion and their stability implications